

Elsa Elisa Yohana Sianturi

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AI Technology Enthusiast with experience in modelling AI, AI evaluation, machine learning, and LLM-powered applications. Interested in AI research, model evaluation, retrieval-augmented generation (RAG), and emerging AI technologies, with a strong habit of exploring research papers, experimenting with new models, and studying methodologies behind modern AI systems. Particularly interested in model reasoning, hallucination analysis, explainability, and responsible AI development

EDUCATION

Bachelor's Degree in Informatics Engineering | Institut Teknologi Sumatera

Aug 2022 - May 2026

Lampung, Indonesia

- GPA: 3.63/4.00
- Focus: Deep Learning, Machine Learning, NLP, and Computer Vision
- Relevant Coursework: Artificial Intelligence, Machine Learning, Deep Learning, Natural Language Processing, Multimodal Machine Learning, Data Visualization and Information

EXPERIENCE

PT Pertamina Hulu Rokan - AI Engineer Intern

Jun 2025 - Jul 2025

Pekanbaru, Indonesia

- Researched and evaluated Retrieval-Augmented Generation (RAG) approaches for enterprise document retrieval
- Integrated LLM-based retrieval and document processing pipelines to enable contextual question answering from PDF land agreement documents
- Evaluated generated answers against original PDF source content to verify grounding, identify unsupported responses, and improve answer reliability, achieving 73.33% answer accuracy in a RAG-based document QA prototype

UPA TIK ITERA - Teaching Assistant

Aug 2023 - Present

Lampung, Indonesia

- **Operating Systems**
Facilitated hands-on laboratory sessions focusing on process management, memory allocation, and operating system fundamentals
- **Introduction to Computing & Software I (PKS I)**
Guided students in procedural programming, computational thinking, and fundamental programming logic
Credential ID: B/5359/IT9.E2/TA.00.00/2024
- **Interaction Design**
Reviewed student assignments and provided structured feedback on clarity, and project quality
Credential ID: 33133/IT9.3.3/TA.00.00/2025

PROJECTS

Multimodal Deep Learning for Self-Harm Meme Detection | [GitHub]

Aug 2025 - Apr 2026

Tech: CLIP, ELECTRA, Gradio

- Built and curated a custom multimodal dataset by collecting, cleaning, and annotating image-text samples using LLM-assisted labeling based on predefined criteria
- Reviewed and validated annotation results to ensure labeling consistency and data reliability
- Conducted model evaluation and error analysis across multimodal and single-modality baselines, achieving 96.1% F1-score

Multimodal City Branding Analysis | [GitHub]

May 2026 - Present

Tech: Image-Text Representation Learning, Social Media Analysis

- Developed a multimodal deep learning model to analyze South Lampung tourism image and caption data from Instagram
- Achieved approximately 89.2% validation accuracy at the selected best epoch based on validation loss
- Collaborated with researchers and students from Urban and Regional Planning and Informatics Engineering

Tech: InsightFace ArcFace, Scikit-learn, OpenCV, SVM, Jupyter Notebook

- Developed an end-to-end facial identification system to automate the recognition of 70 student identities from uploaded facial images
- Engineered a pipeline using offline image augmentation, 512-dimensional InsightFace ArcFace Buffalo_L embeddings, and an RBF-kernel SVM classifier with leakage-resistant Stratified Group K-Fold evaluation
- Processed 1,946 facial samples and achieved 82.0% accuracy, 85.4% weighted precision, and 82.4% weighted F1-score

LEADERSHIP & ORGANIZATIONS

Head of Academic & Scholarship Division — HMIF ITERA

Jan 2025 – Dec 2025

Lampung, Indonesia

- Led 4 academic and scholarship-related programs, served as a speaker in a scholarship-sharing session, and managed mentoring initiatives that contributed to 2 students receiving the Pertamina Sobat Bumi Scholarship

Secretary of Minister of Information Technology — KM ITERA

Apr 2024 – Dec 2024

Lampung, Indonesia

- Managed meeting notes, project documentation, and GitHub-based timeline tracking to coordinate IT-related student government web projects and report progress to the Minister of Information Technology

Head of Entrepreneurship Department — Sobat Bumi Lampung

Oct 2025 – Present

Lampung, Indonesia

- Coordinated vendor communication and production planning for Sobat Bumi Lampung merchandise, including jackets, vests, and community attributes, while supporting entrepreneurship initiatives within the scholarship community

SKILLS

- **Programming & Data Processing:** Python, SQL, Pandas, NumPy
- **AI & Machine Learning:** Machine Learning, Deep Learning, PyTorch, Scikit-learn, Hugging Face, Computer Vision, NLP, Model Evaluation
- **LLM:** RAG, LLM Evaluation, Prompt Engineering, Hallucination Analysis, AI Response Evaluation
- **AI Tools & Frameworks:** OpenCV, LangChain, FAISS
- **Data Analysis & Visualization:** Data Analysis, Data Visualization, Matplotlib, Power BI
- **Research & AI Evaluation:** Literature Review, Experimental Design, Error Analysis, Data Annotation, Data Quality Review, WB

CERTIFICATS

- **AWS re/Start Cloud Computing Program** - Orbit Future Academy & Amazon Web Services (Mar 2026 - Present)
- **Belajar Pemrograman Prosedural dengan Python** - Dicoding (2024) | *Credential ID: 2VX3O4W23ZYQ*
- **Memulai Pemrograman dengan Python** - Dicoding (2024) | *Credential ID: L4PQQ8O2PO01*
- **Belajar Dasar Structured Query Language (SQL)** - Dicoding (2024) | *Credential ID: NVP77M3WVPR0*
- **AI Praktis untuk Productivity** - Dicoding (2024) | *Credential ID: 1OP82QOYLPQK*

AWARDS & HONORS

- **Pertamina Sobat Bumi Scholarship Recipient** - Pertamina Foundation (2024)
- **2nd Winner of UI/UX Competition Point Project 2.0** - HMIF ITERA (2024) | *Credential ID: 313/525/SRT/HMIF/XII/2024*

LANGUAGES & PUBLICATIONS

- **Indonesian:** Native
- **English:** Professional Working Proficiency
- **Publications:** Sianturi, E. E. Y., et al. "Transformasi Digital Layanan Posyandu." PADIMAS, 2025 | [Full Paper]